

Class 13

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Table of contents

Importing data: Setup	1
Toy differential gene expression	5
Setting up for DESeq	10
Volcano Plot	12
Annotations DBI	14
Pathway Analysis	16
Saving our annotated results	19

Importing data: Setup

```
library(BiocManager)
```

```
Bioconductor version '3.22' is out-of-date; the current release version '3.23'
  is available with R version '4.6'; see https://bioconductor.org/install
```

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")
head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	723	486	904	445	1170
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2

	SRR1039517	SRR1039520	SRR1039521
ENSG000000000003	1097	806	604
ENSG000000000005	0	0	0
ENSG000000000419	781	417	509
ENSG000000000457	447	330	324
ENSG000000000460	94	102	74
ENSG000000000938	0	0	0

```
head(metadata)
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871

Q1. There are 38694 genes in the dataset.

```
nrow(counts)
```

```
[1] 38694
```

Q2. There are 4 control cell lines

```
table(metadata$dex)
```

```
control treated
      4      4
```

Toy differential gene expression

```
control.inds <- metadata$dex == "control"
control.counts<- counts[ ,control.inds]
control.mean <- rowMeans(control.counts)
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
      900.75          0.00          520.50          339.75          97.25
ENSG000000000938
      0.75
```

```
#no new functions here
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:Biobase':

combine

The following object is masked from 'package:matrixStats':

count

The following objects are masked from 'package:GenomicRanges':

intersect, setdiff, union

The following object is masked from 'package:Seqinfo':

intersect

The following objects are masked from 'package:IRanges':

collapse, desc, intersect, setdiff, slice, union

The following objects are masked from 'package:S4Vectors':

first, intersect, rename, setdiff, setequal, union

The following objects are masked from 'package:BiocGenerics':

combine, intersect, setdiff, setequal, union

The following object is masked from 'package:generics':

explain

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
control <- metadata %>% filter(dex=="control") #pipe usage from dplyr allows for
#streamlined dataframe modification and processing
control.counts <- counts %>% select(control$id)
control.mean <- rowSums(control.counts)/4
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
          900.75           0.00           520.50           339.75           97.25
ENSG0000000000938
          0.75
```

Q3. You can make the code above more robust by changing the hardcoded 4, to an adaptable version to count the number of control values:

```
control.mean <- rowSums(control.counts)/sum(metadata$dex=="control")
#instead of
control.mean <- rowSums(control.counts)/4
```

Q4.

```
treated <- metadata[metadata[, "dex"]=="treated",]
treated.mean <- rowSums( counts[ ,treated$id] )/sum(metadata$dex=="treated")
head(treated.mean)
```

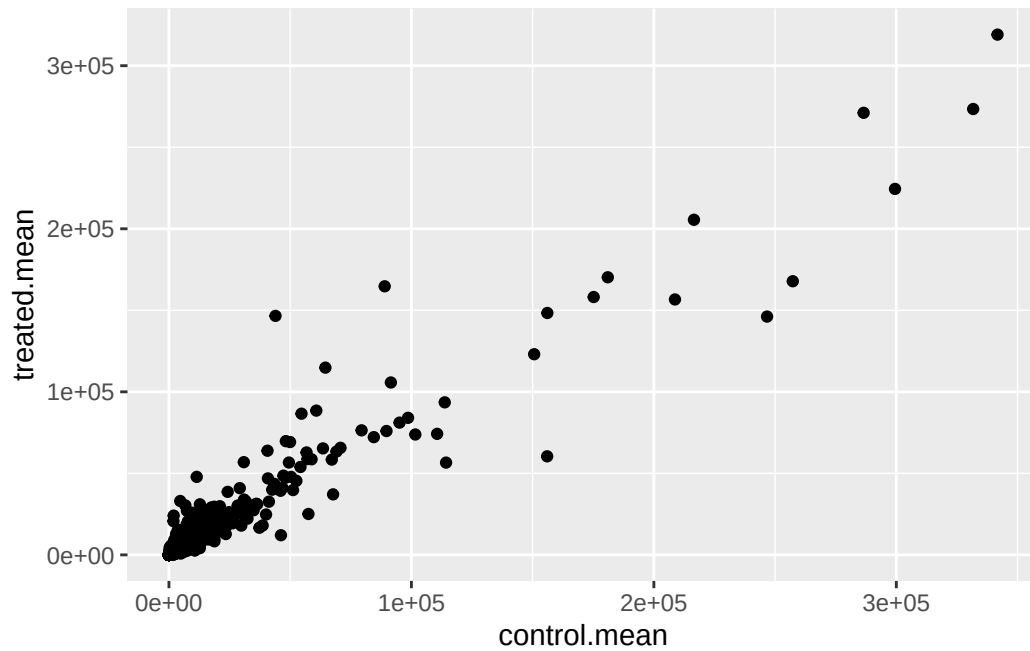
```
ENSG000000000003 ENSG000000000005 ENSG0000000000419 ENSG0000000000457 ENSG0000000000460
                658.00                0.00                546.00                316.50                78.75
ENSG0000000000938
                0.00
```

Combining meancount data:

```
meancounts <- data.frame(control.mean, treated.mean)
head(meancounts)
```

	control.mean	treated.mean
ENSG000000000003	900.75	658.00
ENSG000000000005	0.00	0.00
ENSG0000000000419	520.50	546.00
ENSG0000000000457	339.75	316.50
ENSG0000000000460	97.25	78.75
ENSG0000000000938	0.75	0.00

```
library(ggplot2)
ggplot(meancounts, aes(y = treated.mean, x = control.mean)) +
  geom_point()
```



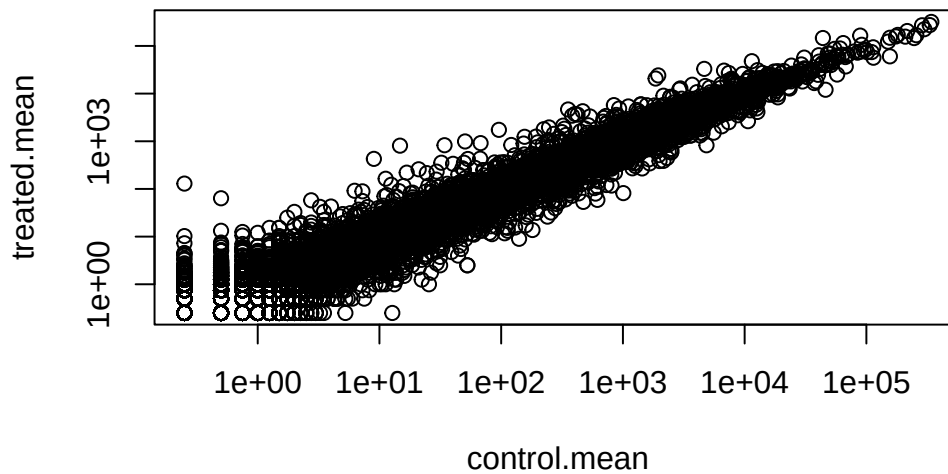
Q5. `Geom_point()` is the function used to generate the dot graph.

Q6. `log="xy"` is the argument to add to `plot()`

```
plot(meancounts, log="xy")
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted  
from logarithmic plot
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted  
from logarithmic plot
```

```
meancounts$log2fc <- log2(meancounts[, "treated.mean"] / meancounts[, "control.mean"])
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

```
zero.vals <- which(meancounts[, 1:2] == 0, arr.ind=TRUE) #which applies a boolean to each value
# which rows satisfy the condition
```

```
to.rm <- unique(zero.vals[, 1]) #returns a vector of unique values
mycounts <- meancounts[-to.rm,]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805

ENSG00000000460	97.25	78.75	-0.30441833
ENSG00000000971	5219.00	6687.50	0.35769358
ENSG00000001036	2327.00	1785.75	-0.38194109

Q7. The purpose of `arr.ind` in the `which()` function is to return the row and column indices for where there are the boolean true values, this would be necessary to avoid extracting zero values, which we want to avoid. Calling `unique` will then make sure there is no redundancy in counting of identical row samples.

```
up.ind <- mycounts$log2fc > 2
down.ind <- mycounts$log2fc < (-2)
```

Q8. There are 250 up-regulated genes greater than 2-fc level

```
sum(up.ind)
```

```
[1] 250
```

Q9. There are 367 down-regulated genes.

```
sum(down.ind)
```

```
[1] 367
```

Q10. No, these results should not be trusted. The analysis only considers fold change magnitude, but a large fold change by itself doesn't tell us whether the difference is statistically significant. We have no p-values here, so we have no way of knowing if these expression differences are statistically significant. We need a proper statistical processing with DESeq before drawing any real conclusions about differential expression.

Setting up for DESeq

```
library(DESeq2)
citation("DESeq2")
```

To cite package 'DESeq2' in publications use:

Love, M.I., Huber, W., Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2 Genome Biology 15(12):550 (2014)

A BibTeX entry for LaTeX users is

```
@Article{,
  title = {Moderated estimation of fold change and dispersion for RNA-
seq data with DESeq2},
  author = {Michael I. Love and Wolfgang Huber and Simon Anders},
  year = {2014},
  journal = {Genome Biology},
  doi = {10.1186/s13059-014-0550-8},
  volume = {15},
  issue = {12},
  pages = {550},
}
```

```
dds <- DESeqDataSetFromMatrix(countData=counts, #runs a DESeq dataset analysis
                              colData=metadata,
                              design=~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds
```

```
class: DESeqDataSet
dim: 38694 8
metadata(1): version
assays(1): counts
rownames(38694): ENSG000000000003 ENSG000000000005 ... ENSG00000283120
               ENSG00000283123
rowData names(0):
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(4): id dex celltype geo_id
```

```
#countData are gene counts across different experiments
#colData is our metadata about those count columns
```

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds) #results of the DESeq analysis
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

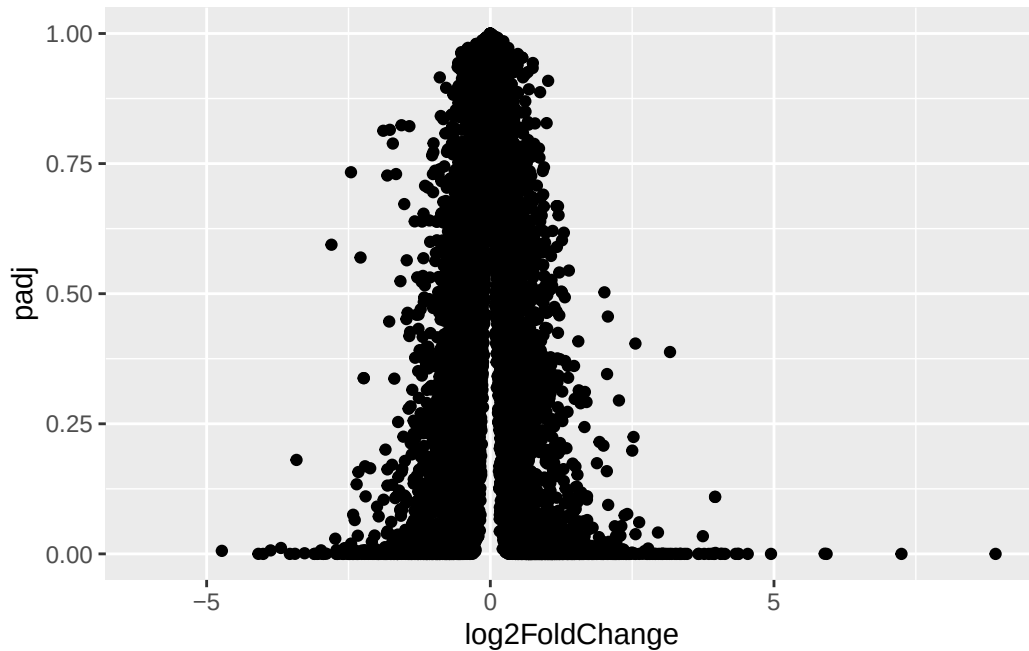
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG0000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG0000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG0000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG0000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG000000000003	0.163017				
ENSG000000000005	NA				
ENSG0000000000419	0.175937				
ENSG0000000000457	0.961682				
ENSG0000000000460	0.815805				
ENSG0000000000938	NA				

Volcano Plot

```
library(ggplot2)
ggplot(res) +
  aes(x = log2FoldChange, y = padj) +
  geom_point()
```

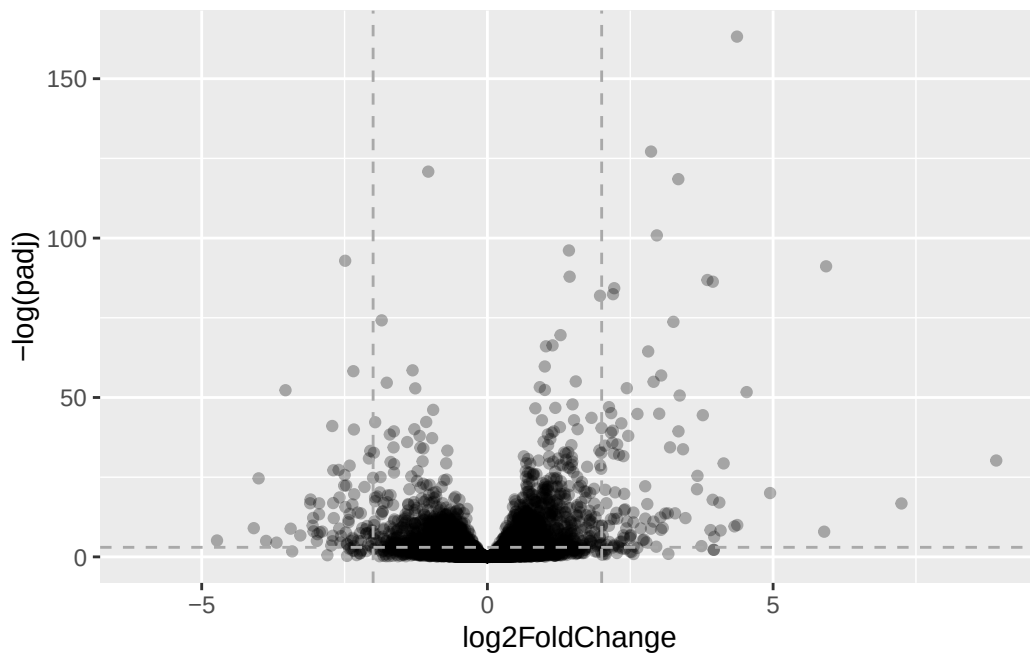
Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).



That plot is not very useful because we do not care about plots with extremely high p-values

```
library(ggplot2)
ggplot(res) +
  aes(x = log2FoldChange, y = -log(padj)) + #log2FoldChange applies a logarithm scale to a variable
  geom_point(alpha = 0.3) +
  geom_hline(yintercept=-log(0.05), col = "darkgray", linetype="dashed") + #vline and hline are used to create
  #of a horizontal and vertical line in order to show critical cutoff values
  geom_vline(xintercept=c(-2,2), col="darkgray", linetype="dashed")
```

Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).



Annotations DBI

Q11. Completing the code below, adding “GENENAME” and “ENTREZID” and “UNIPROT”

```
library("AnnotationDbi")
```

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

```
select
```

```
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

[1] "ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6] "ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11] "GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"

```
[16] "OMIM"          "ONTOLOGY"      "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"          "PROSITE"       "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="ENTREZID",
                     keytype="ENSEMBL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$uniprot <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="UNIPROT",
                     keytype="ENSEMBL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="GENENAME",
                      keytype="ENSEMBL",
                      multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
ENSG000000000971	5760.148362	0.459282	0.234318	1.960081	0.0499864
ENSG000000001036	2025.391794	-0.228244	0.124999	-1.825965	0.0678555
ENSG000000001084	652.173308	-0.253031	0.202576	-1.249064	0.2116417

ENSG00000001167	411.684395	-0.533733	0.228792	-2.332831	0.0196570
	padj	entrez	uniprot	genename	
	<numeric>	<character>	<character>	<character>	
ENSG000000000003	0.163017	7105	AOA087WYV6	tetraspanin 6	
ENSG000000000005	NA	64102	Q9H2S6	tenomodulin	
ENSG000000000419	0.175937	8813	H0Y368	dolichyl-phosphate m..	
ENSG000000000457	0.961682	57147	X6RHX1	SCY1 like pseudokina..	
ENSG000000000460	0.815805	55732	A6NFP1	FIGNL1 interacting r..	
ENSG000000000938	NA	2268	B7Z6W7	FGR proto-oncogene, ..	
ENSG000000000971	0.200967	3075	A5PL14	complement factor H	
ENSG00000001036	0.246740	2519	E9PEB6	alpha-L-fucosidase 2	
ENSG00000001084	0.495029	2729	AOA2R8YEL6	glutamate-cysteine l..	
ENSG00000001167	0.105242	4800	P23511	nuclear transcriptio..	

Pathway Analysis

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
```

```
data(kegg.sets.hs)
```

```
# Examine the first 2 pathways in this kegg set for humans
head(kegg.sets.hs, 2)
```



```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"  
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"  
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"  
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"  
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"  
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"  
[49] "8824" "8833" "9" "978"
```

```
foldchanges = res$log2FoldChange  
names(foldchanges) = res$entrez  
head(foldchanges)
```

```
          7105          64102          8813          57147          55732          2268  
-0.35070296          NA  0.20610728  0.02452701 -0.14714263 -1.73228897
```

```
# Get the results  
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names  
[1] "greater" "less" "stats"
```

```
# Look at the first three down (less) pathways  
head(keggres$less, 3)
```

	p.geomean	stat.mean	p.val
hsa05332 Graft-versus-host disease	0.0004250607	-3.473335	0.0004250607
hsa04940 Type I diabetes mellitus	0.0017820379	-3.002350	0.0017820379
hsa05310 Asthma	0.0020046180	-3.009045	0.0020046180

	q.val	set.size	exp1
hsa05332 Graft-versus-host disease	0.09053792	40	0.0004250607
hsa04940 Type I diabetes mellitus	0.14232788	42	0.0017820379
hsa05310 Asthma	0.14232788	29	0.0020046180

```
pathview(gene.data=foldchanges, pathway.id="hsa05310")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/derke/Desktop/BIMM 143/Class 13

Info: Writing image file hsa05310.pathview.png

Q12. The top 2 down-regulated pathways from `keggres$less` are hsa05332 (Graft-versus-host disease) and hsa04940 (Type I diabetes mellitus).

```
# Pathview figures for top 2 down-regulated pathways
pathview(gene.data=foldchanges, pathway.id="hsa05332")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/derke/Desktop/BIMM 143/Class 13

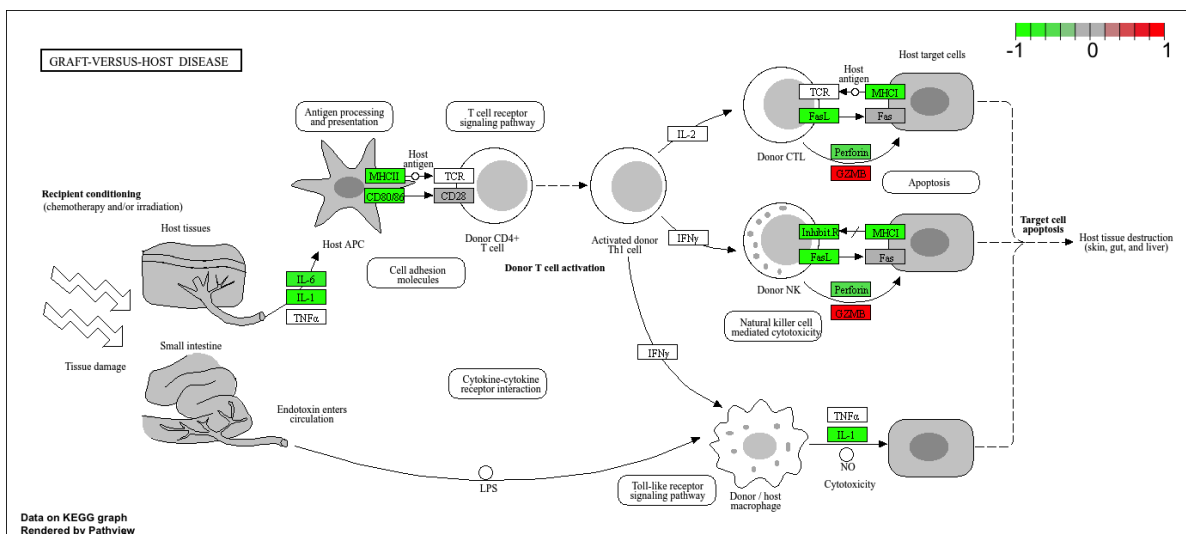
Info: Writing image file hsa05332.pathview.png

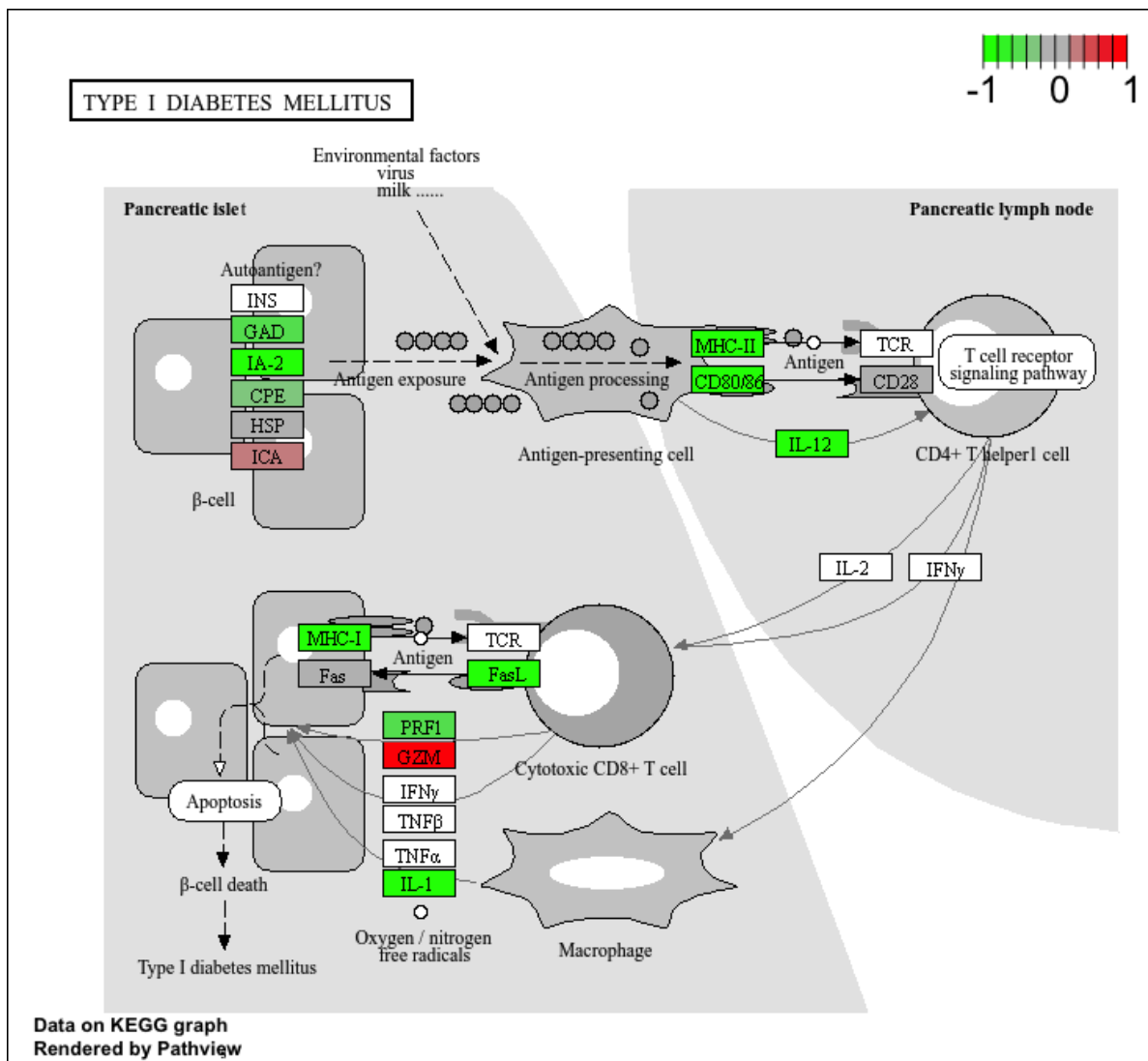
```
pathview(gene.data=foldchanges, pathway.id="hsa04940")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/derke/Desktop/BIMM 143/Class 13

Info: Writing image file hsa04940.pathview.png





Saving our annotated results

```
write.csv(res, file="myresults_annotated.csv")
```